Lec 34 : Rotations in QM

SAN 3.1 – 3.3

$$|\alpha\rangle_R = \mathcal{D}(R)\,|\alpha\rangle$$

How do expectation values change?

$$\langle \hat{O}\rangle_\alpha = \langle\alpha|\hat{O}|\alpha\rangle \longrightarrow {}_R\langle\alpha|\hat{O}|\alpha\rangle_R$$

$$= \langle\alpha|\mathcal{D}^\dagger(R)\,\hat{O}\,\mathcal{D}(R)|\alpha\rangle$$

Let's look at angular momentum itself:

$$\langle \hat{J}_x\rangle_\alpha \longrightarrow \langle\alpha|\mathcal{D}^\dagger(R)\,\hat{J}_x\,\hat{\mathcal{D}}(R)|\alpha\rangle$$

$$e^{\frac{i}{\hbar}\hat{J}_z\theta}\,\hat{J}_x\,e^{-\frac{i}{\hbar}\hat{J}_z\theta} = \left(1+\tfrac{i}{\hbar}\hat{J}_z\theta+\dots\right)\hat{J}_x\left(1-\tfrac{i}{\hbar}\hat{J}_z\theta+\dots\right)$$

$$= \left(\hat{J}_x + \underbrace{\tfrac{i\theta}{\hbar}[\hat{J}_z,\hat{J}_x]}_{i\hbar\hat{J}_y} + \tfrac{1}{2}\left(\tfrac{i\theta}{\hbar}\right)^2\underbrace{[\hat{J}_z,[\hat{J}_z,\hat{J}_x]]}_{i\hbar\hat{J}_y\ \to\ i\hbar(-i\hbar\hat{J}_x)} + \dots\right)$$

• Baker Hausdorff formula.

$$= \hat{J}_x\left(1-\frac{\theta^2}{2!}+\dots\right) - \hat{J}_y\left(\theta+\frac{\theta^3}{3!}+\dots\right)$$

$$= \hat{J}_x\cos\theta - \hat{J}_y\sin\theta$$

$$\langle \hat{J}_x\rangle_\alpha \longrightarrow \langle\alpha|\hat{\mathcal{D}}^\dagger(\hat{z},\theta)\,\hat{J}_x\,\hat{\mathcal{D}}(z,\theta)|\alpha\rangle$$

$$= \langle\hat{J}_x\rangle_\alpha\cos\theta - \langle\hat{J}_y\rangle_\alpha\sin\theta.$$

In general: $\langle J_i\rangle = R_{ij}\langle J_j\rangle$ ⇒ Defines a Vector Operator

For infinitesimal operation:

$$\hat{\mathcal{D}}^\dagger(\hat{n},d\theta)\,\hat{V}_i\,\hat{\mathcal{D}}(\hat{n},d\theta) = (R_{\hat{n}})_{ij}\hat{V}_j \quad \text{where}$$

$$(R_{\hat{n}}) = \delta_{ij} - i(G\cdot\hat{n})_{ij}\,d\theta$$

Recall we had $(G_k)_{ij} = -i\varepsilon_{ijk}$; $(R_k)_{ij} = \delta_{ij} - \varepsilon_{ijk}d\theta$

So the definition of a vector being a vector is equivalent to: $\sum_k(\varepsilon_{ijk}\hat{n}_k)$

$$\left(\mathbb{1} + \frac{i\,d\theta}{\hbar}(\hat{n}\cdot\hat{\vec{J}})\right)\hat{V}_i\left(\mathbb{1} - \frac{i\,d\theta}{\hbar}(\hat{n}\cdot\hat{\vec{J}})\right) = \left(\delta_{ij} - d\theta(\varepsilon\cdot\hat{n})_{ij}\right)\hat{V}_j$$

$$\hat{V}_i + \frac{i\,d\theta}{\hbar}\hat{n}\cdot[\hat{\vec{J}}, \hat{V}_i] + O(d\theta^2) = \hat{V}_i - d\theta(\varepsilon\cdot\hat{n})_{ij}V_j$$

$$\Rightarrow \quad \sum_k n_k[\hat{J}_k, \hat{V}_i] = \sum_k(i\hbar\varepsilon_{ijk}V_j)\cdot\hat{n}_k$$

$$\Rightarrow \quad \boxed{[\hat{V}_i, \hat{J}_j] = i\hbar\varepsilon_{ijk}\hat{V}_k} \;//$$

↳ Defines a vector. Clearly $\hat{J}_i$ is a vector because $[\hat{J}_i, \hat{J}_j] = i\hbar\varepsilon_{ijk}\hat{J}_k$

Question: Show that $\hat{x}$ and $\hat{p}$ are vectors by using $\hat{J} = \hat{L} = \hat{x}\times\hat{p}$. (recal $(\hat{A}\times\hat{B})_k = \varepsilon_{ijk}\hat{A}_i\hat{B}_j$)

Answer:
$$[\hat{x}_i, \varepsilon_{lmj}\hat{x}_l\hat{p}_m] = \varepsilon_{lmj}\hat{x}_l[\hat{x}_i, \hat{p}_j]$$
$$= \varepsilon_{lmj}\hat{x}_l\,i\hbar\delta_{ij}$$
$$= i\hbar\varepsilon_{lij}\hat{x}_l = i\hbar\varepsilon_{ijk}\hat{x}_k$$

and similarly for $\hat{p}$.

BTW: If $[D(R), \hat{V}] = 0$, then $\hat{V}$ is said to be rotationally invariant.

$D^\dagger\hat{V}_iD = R_{ij}V_j \longrightarrow \hat{V}_i = R_{ij}\hat{V}_j$ stays the same!

Two State system again.

Whatever generates rotations in $\mathcal{H}$ space we work in.

$\hat{S}_x, \hat{S}_y, \hat{S}_z$ spin operators: $\hat{J} \to \hat{S}$

Recall spin operators satisfy $[\hat{S}_x, \hat{S}_y] = i\hbar \hat{S}_z$

The definition of a vector was equivalent to our commutation relation, so:

$$D^\dagger(\hat{n}, \theta)\, \hat{S}_i\, D(\hat{n}, \theta) = (R\hat{n})_{ij}\, \hat{S}_j$$

Note our Hilbert space is just $\mathbb{C}^2$, but if we had a 2-state system in 3D space, then $\mathcal{H} = \mathbb{R}^3 \otimes \mathbb{C}^2$ and $\hat{J} = \hat{L} + \hat{S}$.

$\hat{L}$ and $\hat{s}$ act on different parts of our states:

$\hat{J}|\vec{x}, s\rangle \sim \hat{L}|x\rangle \otimes \hat{s}|s\rangle$.

For example, for a rotation about the $\hat{z}$ axis:

$$\begin{cases} \langle \hat{S}_x \rangle \to \langle \hat{S}_x \rangle \cos\theta - \langle \hat{S}_y \rangle \sin\theta \\ \langle \hat{S}_y \rangle \to \langle \hat{S}_x \rangle \sin\theta + \langle \hat{S}_y \rangle \cos\theta \\ \langle \hat{S}_z \rangle \to \langle \hat{S}_z \rangle \quad \left(\text{n.b. } [D(\hat{z}, \theta), \hat{S}_z] = 0 \right) \end{cases}$$

Now in terms of the kets: $\hat{S}_z = \frac{\hbar}{2}\left(|+\rangle\langle +| - |-\rangle\langle -|\right)$

So $\hat{D}(\hat{z}, \theta) = e^{-\frac{i\hat{S}_z}{\hbar}\theta} = \left(1 - \frac{i\theta}{2}\left(|+\rangle\langle +| - |-\rangle\langle -|\right) + \left(\frac{-i}{\hbar}\theta\right)^2 \frac{(|+\rangle\langle +| - |-\rangle\langle -|)}{4} + \cdots\right)$

$$\hat{D}(\hat{z},\theta) = e^{\frac{-i\theta}{2}}|+\rangle\langle+| + e^{\frac{+i\theta}{2}}|-\rangle\langle-|$$

dyad form of rotation around $\hat{z}$ axis in 2-state system.

⇒ Any given state can be represented as:

$$|\alpha\rangle = |+\rangle\langle+|\alpha\rangle + |-\rangle\langle-|\alpha\rangle = c_+|+\rangle + c_-|-\rangle$$

After rotation through $\hat{z}$ axis:

$$|\alpha\rangle_R = \hat{D}(\hat{z},\theta)|\alpha\rangle$$
$$= e^{\frac{-i\theta}{2}}|+\rangle\langle+|\alpha\rangle + e^{\frac{i\theta}{2}}|-\rangle\langle-|\alpha\rangle$$
$$= e^{\frac{-i\theta}{2}}c_+|+\rangle + e^{\frac{i\theta}{2}}c_-|-\rangle$$

Half angle rotation!

$$\hat{D}(\hat{z},2\pi)|\alpha\rangle = e^{-i\pi}c_+|+\rangle + e^{i\pi}c_-|-\rangle$$
$$= -(c_+|+\rangle + c_-|-\rangle)$$
$$= -|\alpha\rangle.$$

Full 2π rotation leads to an overall minus sign.

Spin precession, again!

$\hat{H} = -\vec{\mu}\cdot\vec{B}$ Take $\vec{B} = B\hat{z}$ in $\hat{z}$ axis.

Then with $\vec{\mu} = \gamma\vec{S} = -\frac{e}{m}\vec{S}$

$$\hat{H} = \frac{eB}{m}\hat{S}_z = \omega\hat{S}_z \Rightarrow \text{Time evol.}$$
$$\hat{u}(t) = e^{\frac{-i\hat{H}t}{\hbar}} = e^{\frac{-i\omega t}{\hbar}\hat{S}_z}$$

Compare with rotation operator:

$\hat{\mathcal{U}}(t) = \hat{\mathcal{D}}(\hat{z}, \omega t)$ rotation by (ωt)

The expectation value of spin operator precesses with a period $\frac{2\pi}{\omega}$:

$$\langle \vec{S} \rangle = \langle \alpha | \mathcal{D}^\dagger(\hat{z}, \omega t)\, \vec{S}\, \mathcal{D}(\hat{z}, \omega t) | \alpha \rangle$$

$$= \langle \alpha |\, R_z(\omega t)\, \vec{S}\, | \alpha \rangle$$

$$= \begin{pmatrix} \langle S_x \rangle \cos\omega t - \langle S_y \rangle \sin\omega t \\ \langle S_x \rangle \sin\omega t + \langle S_y \rangle \cos\omega t \\ \langle S_z \rangle \end{pmatrix}$$

Note $\langle \vec{S} \rangle \xrightarrow[\omega t = 2\pi]{} \langle \vec{S} \rangle$

But somehow $|\alpha\rangle \xrightarrow[\omega t = 2\pi]{} -|\alpha\rangle$

On its own, this minus sign is not observable. But in a superposition it is.

Read S&N <u>3.2.3</u> and <u>3.3.1 – 3</u>

Attempt Problem 3.7 in S&N !

